

FINTECH INNOVATION • BIOMETRIC AUTHENTICATION

ThumbPay

Your identity is the only wallet you will never leave behind.

A device-free payment layer built on top of existing global financial rails.

THE PARADIGM SHIFT

The Evolution of Payments

Payments evolved from cash to physical cards, then to smartphones. The final evolutionary step removes hardware dependencies entirely. Identity becomes the payment credential.

The Problem with Modern POS

1. Device & Battery Dependency

Mobile wallets and UPI apps depend heavily on battery life, cell signal, and app stability. 15% of mobile wallet users experience transaction failure due to dead phone batteries.

2. Checkout Queue Friction

Physical card swipes and QR scanning take 12 to 18 seconds per transaction. Retail queue delays cause over \$38 Billion in abandoned store purchases globally every year.

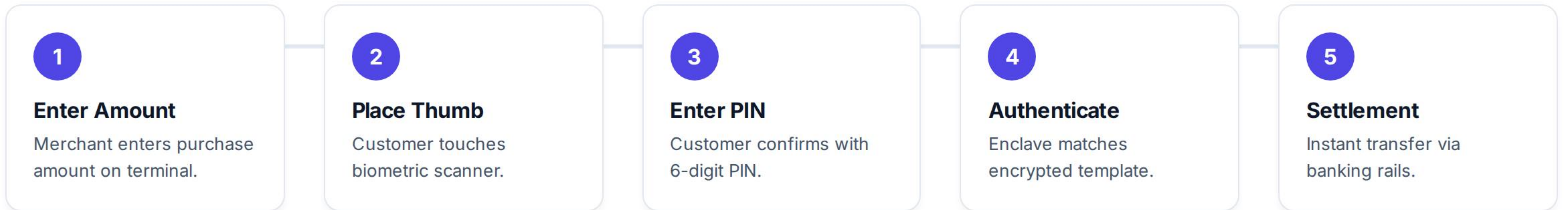
ThumbPay: Identity as Wallet

ThumbPay is an authentication layer operating on top of existing payment networks like UPI, card networks, and banks.

- **Zero Hardware Carried:** No phone, wallet, or card required.
- **Ultra-Fast Settlement:** Transactions complete in under 3 seconds.
- **Complementary Layer:** Integrates with NPCI, UPI, and core banking switches.



5-Step Transaction Journey



Bank-Grade Biometric Security



Zero Raw Fingerprints

Raw images are never stored.
Fingerprints convert immediately into non-reversible mathematical vectors.



Hardware Enclave

Template matching occurs within Secure Element (SE) & Hardware Security Modules (HSM FIPS 140-2 Level 3).



Tokenized Payloads

Every payment request generates dynamic AES-256 tokens paired with a 2-factor encrypted PIN hash.

Market Opportunity & Scale

228B+

Annual Digital Transactions

India Leading Real-Time Payments

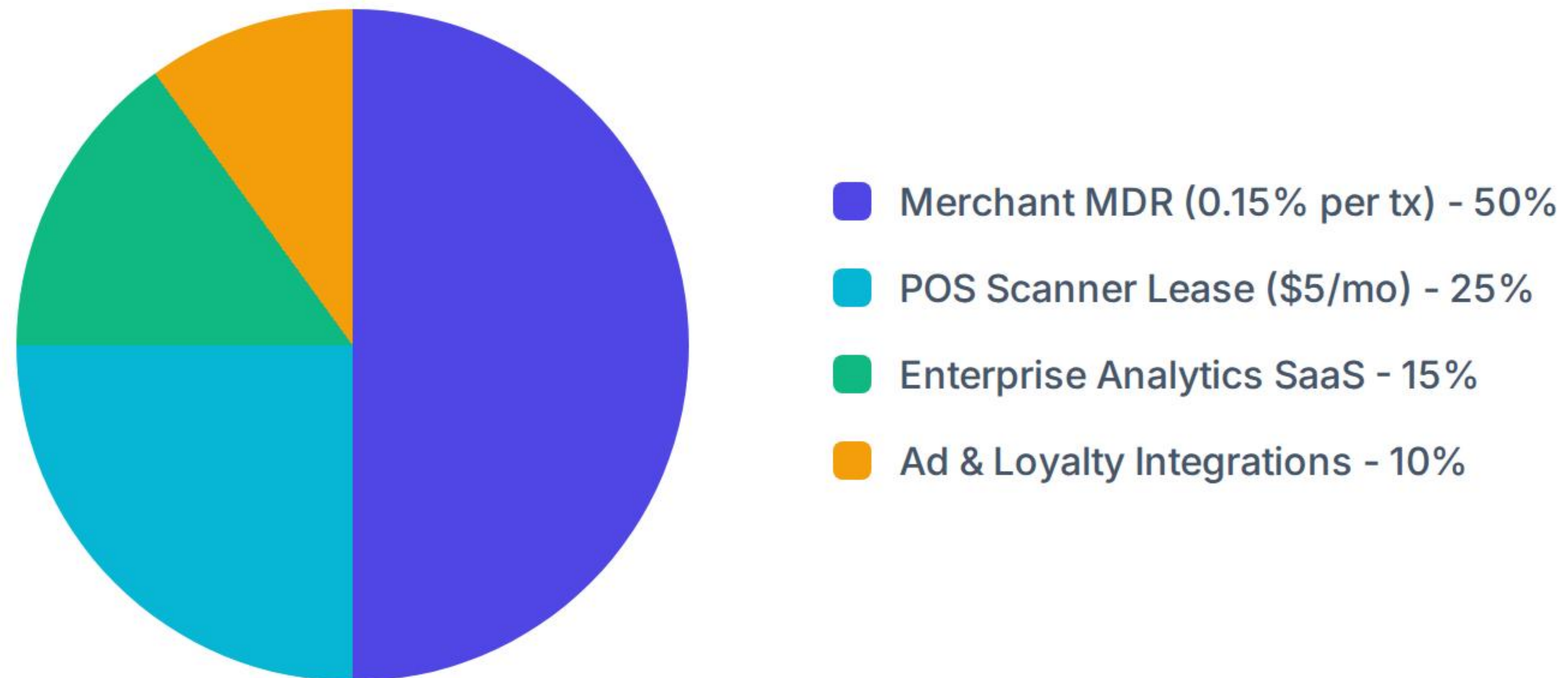
India processed over 228 Billion UPI transactions in 2025 alone, representing nearly 49% of all real-time payments globally.

The global biometric payment market is expanding at a 16.8% CAGR to reach \$25.7 Billion by 2030, presenting a massive market opportunity for hardware-free checkout.

Competitive Landscape

Feature	Mobile UPI / Wallets	Cards (Debit/Credit)	ThumbPay
Hardware Required	Smartphone	Plastic Card	None
Checkout Time	18 Seconds	12 Seconds	< 3 Seconds
Battery / Data Failure	High Risk	No Risk	Zero Risk
Authentication Layer	App Passcode + PIN	Chip / PIN	Biometric + PIN (2FA)

Diversified Monetization



Strategic Target Segments



University Campuses

High density, tech-forward Gen Z users making high-frequency micro-payments at cafeterias & bookstores.



Cafes & Supermarkets

Retail environments plagued by long checkout queues where speed directly increases store revenue.



Transit & Pharmacies

High-throughput transit turnstiles and urgent situations where consumers may lack cash or mobile devices.

Five-Year Financial Model



• Operational break-even projected at Month 28 following launch.

Source: PitchBook Fintech Industry Benchmarks

JOIN US

Join the Payment Revolution

ThumbPay is building the hardware-free payment infrastructure of tomorrow.

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Image Sources



<https://www.poszeo.com/wp-content/uploads/2026/04/Retail-POS-Hardware-How-to-Build-the-Right-Checkout-Stack-for-Real-Store-Deployment-5-1300x727.webp>

Source: www.poszeo.com